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홈 (/rol/app/) (home) > Observing OpenShift Service Mesh > Chapter 1: Observing OpenShift Service Mesh

Observing OpenShift Service Mesh

Lesson

Lab Environment

Guided Exercise: Introduction to Red Hat OpenShift observability and OpenShift Service Mesh

Use the traffic graph to visualize service mesh topology and interactions.

Outcomes

- Use the traffic graph to visualize service mesh topology and interactions.
- Analyze service performance data through different graph types.
- Monitor service health and traffic patterns in real-time.
- Compare OpenShift Service Mesh console plug-in (OSSMC) and Kiali console capabilities for service mesh observability.

As the student user on the workstation machine, use the lab command to prepare your environment for this exercise, and to ensure that all required resources are available.

```
[student@workstation ~]$ lab start meshobservability-intro
```

Instructions

In this exercise, you have access to a Red Hat OpenShift Service Mesh (OSSM) installation in the `istio-system` cluster namespace. The installation includes the OSSM console plug-in (OSSMC) and the `UIPlugin` resource from the Observability operator.

The `meshobservability-intro` OpenShift project contains the Bookinfo application, along with a traffic generator that generates continuous traffic.

1. Access the OpenShift web console to inspect a working mesh.
 - 1.1. On the workstation machine, open the OpenShift web console in your browser.

Go to `https://console-openshift-console.apps.ocp4.example.com`, click `htpasswd_provider` and use `developer` as the user and the password.
 - 1.2. In the Administrator perspective in the OpenShift console click **Service Mesh** → **Traffic Graph**.



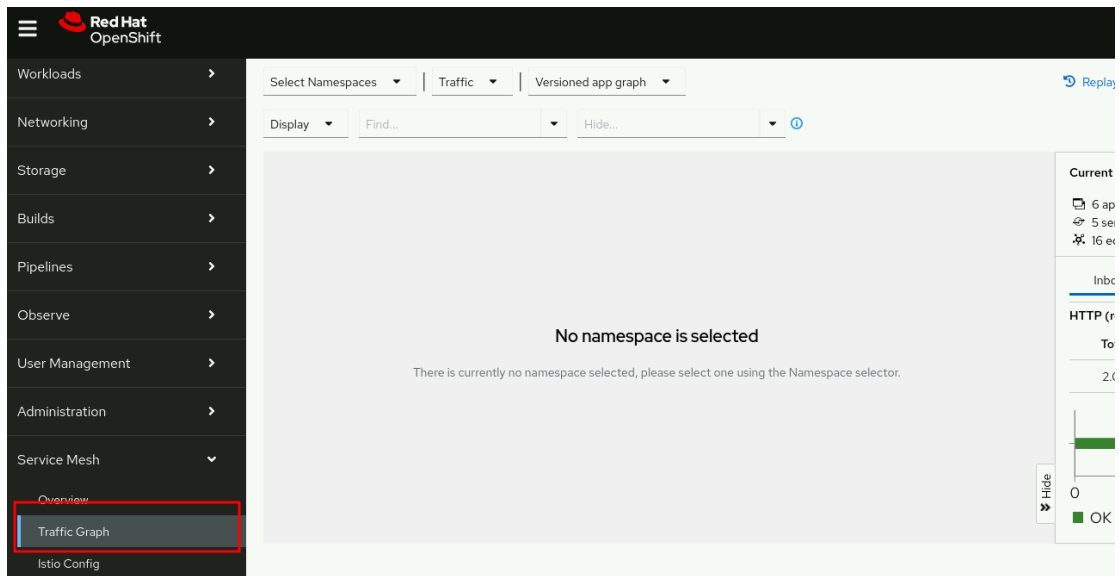


Figure 1.12: Access to the Traffic Graph

- 1.3. Add the meshobservability-intro namespace to the graph by selecting it in **Select Namespaces**. Then, click anywhere outside the text box to apply the selection.

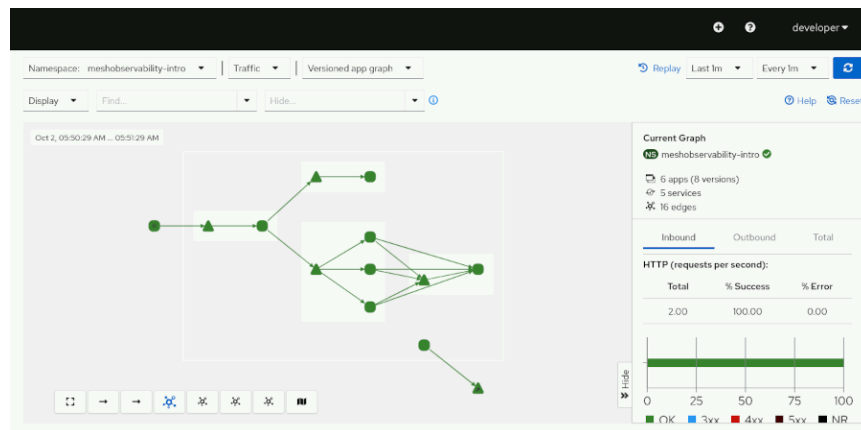


Figure 1.13: Mesh traffic graph overview

- 1.4. Change the graph type from **Versioned app graph** to **Service graph** and inspect the legend.

The service graph type is the simplest. The main entry point of the application is the productpage microservice. The productpage service depends on the other two services, details and reviews. Furthermore, the reviews service is dependent on the ratings service.

Click the legend display icon to show the legend of the graph.

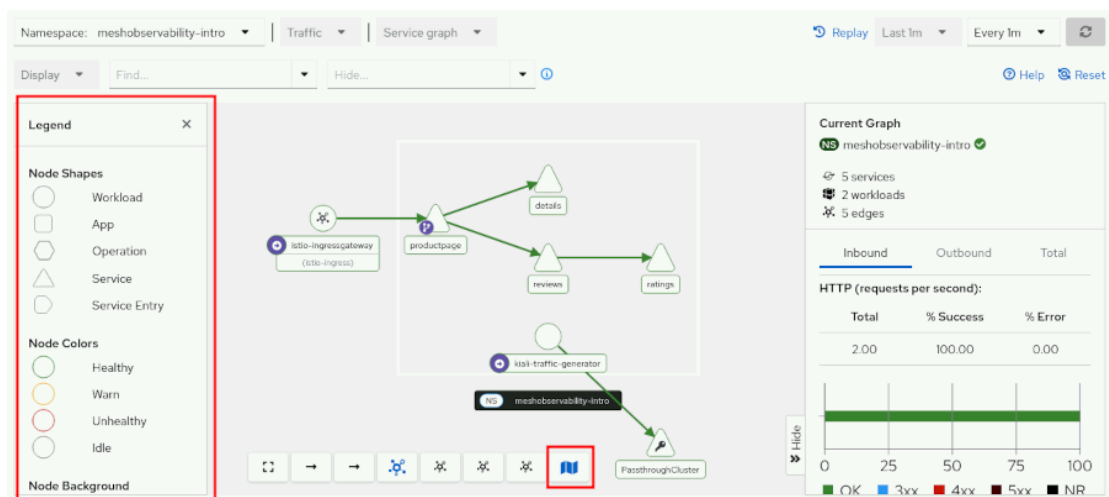


Figure 1.14: The legend side panel

Observe the shapes that represent node types in the mesh, and the meanings of the colors for nodes, edges, and traffic animations. The Bookinfo application has all its nodes and edges in green.

2. Add errors to the Bookinfo application.

2.1. Stop the ratings-v1 deployment to add failures to the Bookinfo application.

In the OpenShift console, go to the **Workloads** → **Deployments** menu, and click **ratings-v1**. Ensure that you select the **meshobservability-intro** project on the **Project** combo box.

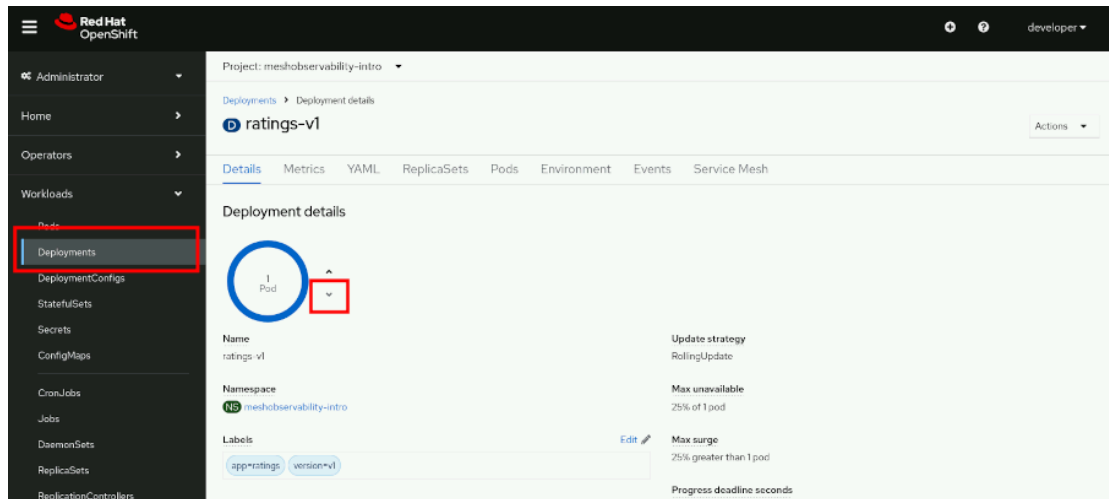


Figure 1.15: The ratings deployment

Then, click the down arrow button to stop the running pod.

2.2. Return to the traffic graph at **Service Mesh** → **Traffic Graph**, and wait two or three minutes to inspect the graph showing failing nodes and edges.

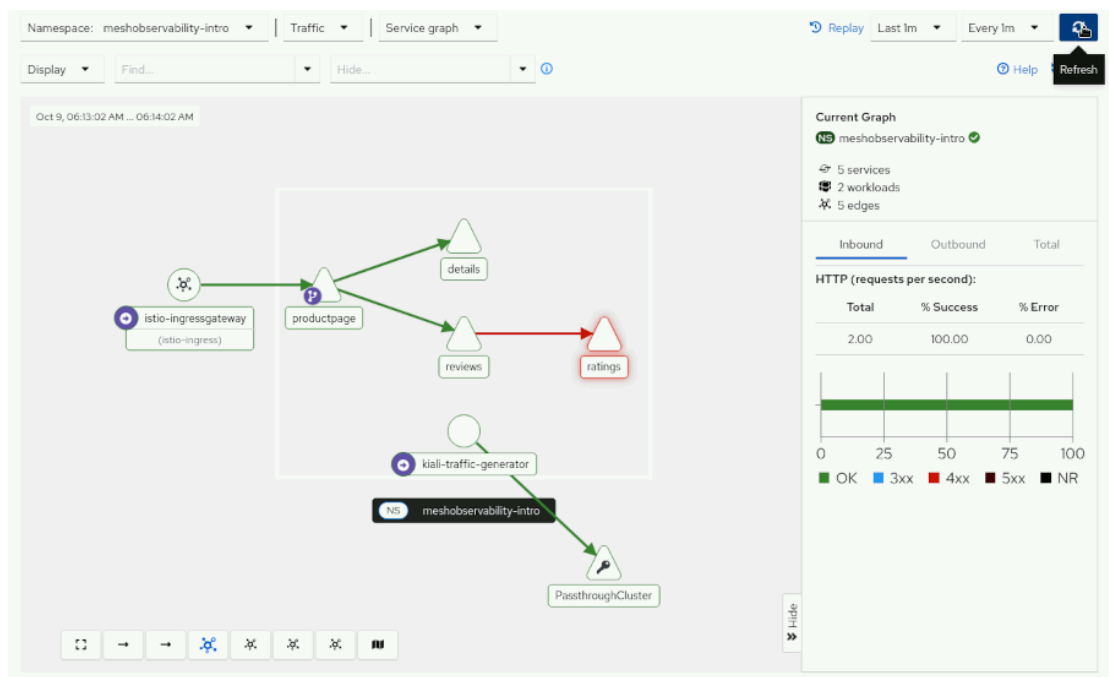


Figure 1.16: Service traffic graph showing errors

2.3. After a few minutes with the service stopped, change the **Traffic metrics per refresh** combo box to **Last 5m**.

Depending on the time range selected and the time spent on the steps, you see different statuses of the mesh over time.

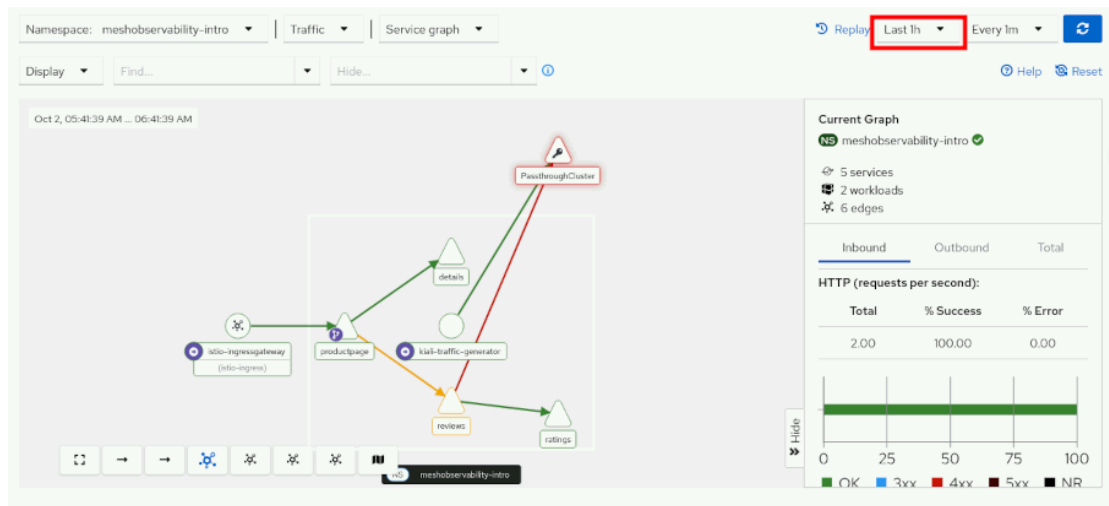


Figure 1.17: Service traffic graph showing degraded status

Test with different time ranges.

- 2.4. Start the ratings-v1 deployment to one pod to restore the Bookinfo application.

Go to the **Workloads** → **Deployments** menu, and click **ratings-v1**. Then, click the up arrow button to start a ratings-v1 deployment's pod.

3. Inspect traffic and performance data in the mesh.

- 3.1. Return to the traffic graph.

In the **Display** combo box, select **Traffic Distribution** and **Traffic Animation**.

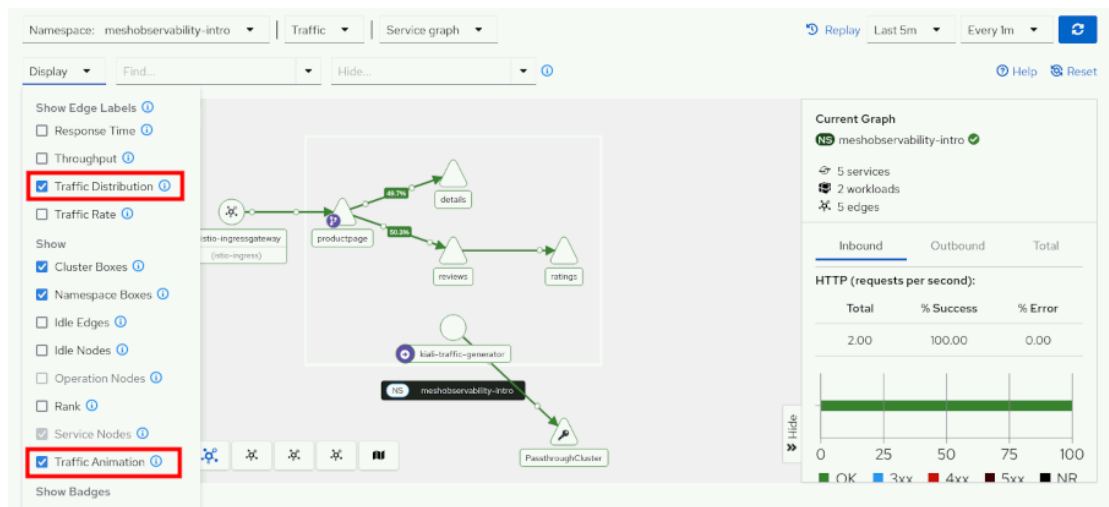


Figure 1.18: Graph showing traffic animation and distribution

Use the scroll wheel on your mouse to increase the zoom level. When you increase the zoom level, the percentage of HTTP traffic on each edge shows.

- 3.2. Change the graph from **Service graph** to **Workload graph** and inspect the traffic distribution of the reviews service, and its three different versions.

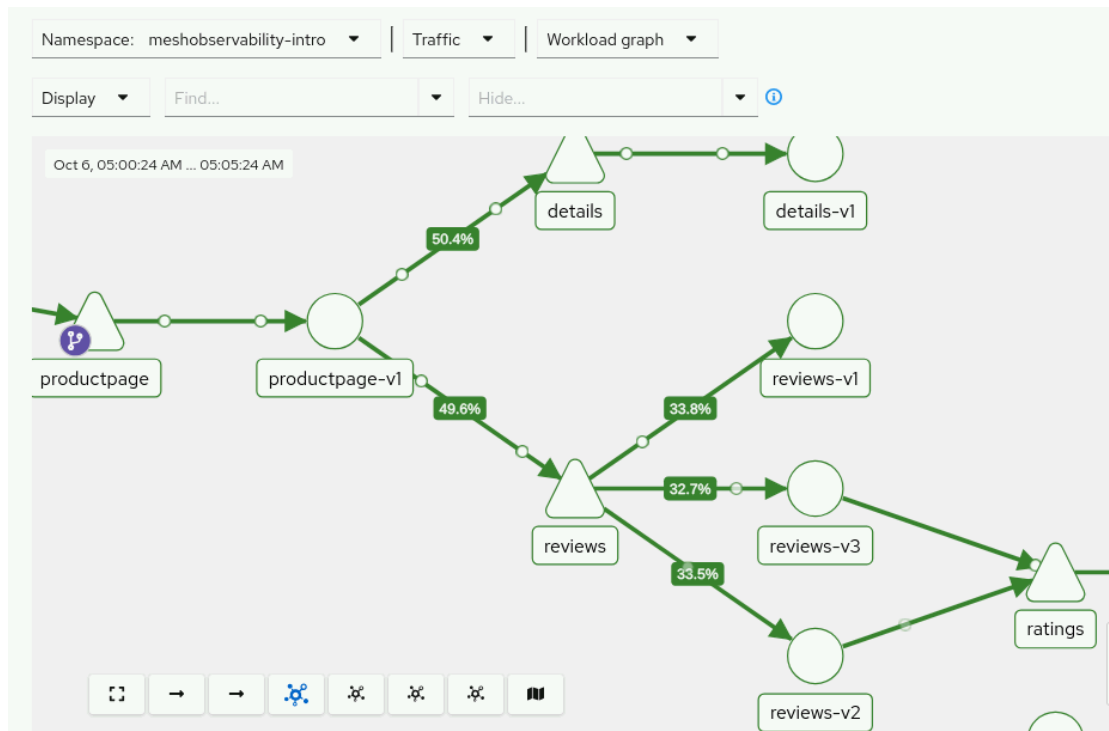


Figure 1.19: Traffic distribution between the three versions of the reviews service

3.3. Deactivate the CPU intensive traffic animation, and inspect performance data.

In the **Display** combo box, unselect **Traffic Animation**, and select **Response Time → 95th Percentile**. This metric shows the response time of 95% of the requests.

The workload graph enables you to infer information even without complete information about the observed services in the mesh. For example, why is the `reviews-v1` service version the fastest of the three versions? The graph shows that version one does not depend on the `ratings` microservice, whereas versions two and three do.

Not all graph types can show all data. For example, if you change to the service graph type, then the response time information does not display.

3.4. Inspect specific data from workloads.

The traffic graph provides quick access to workload details.

For example, right-click the `productpage-v1` workload. Remember from the legend that a circle represents workloads.

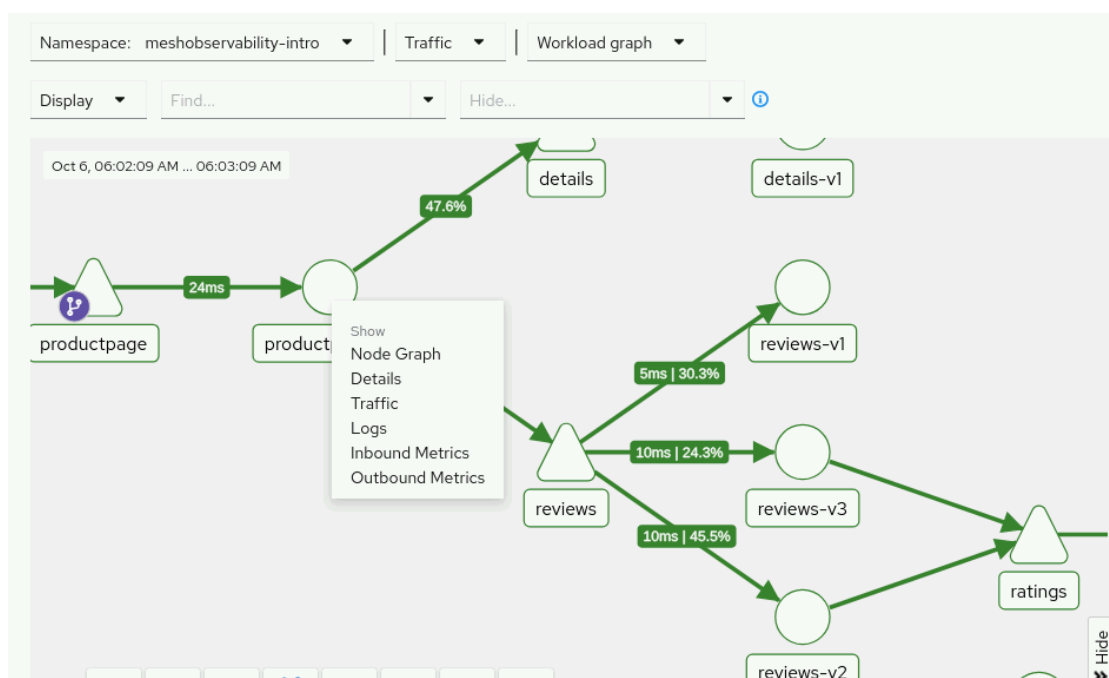


Figure 1.20: Right-click menu on a workload element in the mesh

Click **Details**. This redirects you to the **Service Mesh** tab of the **productpage-v1** Kubernetes deployment object.

Note that you can access the service logs directly from the traffic graph or the inbound and outbound metrics.

3.5. Inspect specific data for services.

Right-click the **reviews** service. Remember from the legend that a triangle represents services.

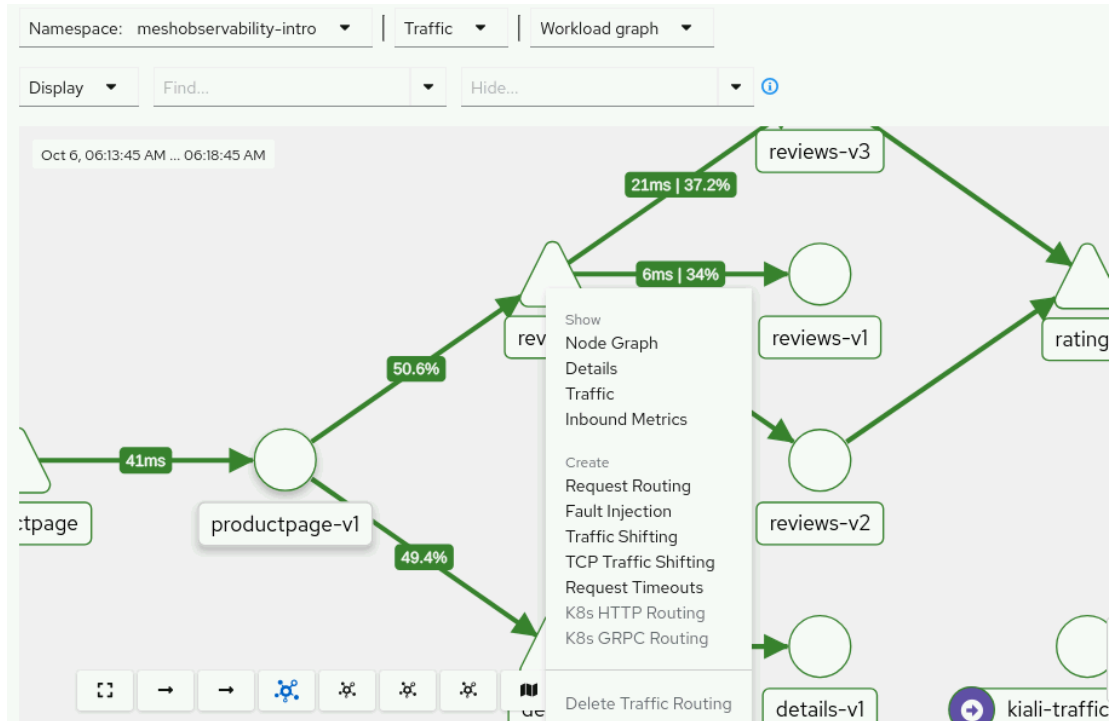


Figure 1.21: Right-click menu on a service element in the mesh

Note that you can create objects to manipulate the traffic of the service, such as fault injections, timeouts, request routing, and others.

Click **Inbound Metrics** to inspect the specific metrics of the **reviews** microservice.

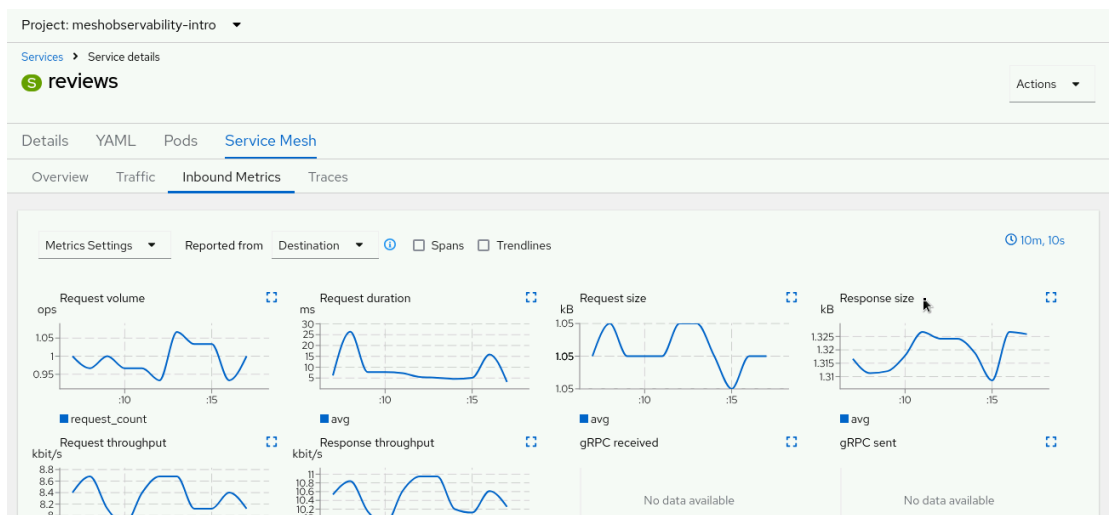


Figure 1.22: Inbound metrics for the reviews service

Remember that the OpenShift console dashboards include a time selector to set the refresh interval and the time range for the dashboard's data.

4. Observe the mesh with the Kiali external console.

- 4.1. From the workstation machine, open a web browser and go to <https://kiali-istio-system.apps.ocp4.example.com>.

4.2. Log in to the Kiali console.

Click `htpasswd_provider` and `developer` as the user and the password.

4.3. Verify that this console shows the same information as the OSSMC plug-in in the OpenShift console:

The **Overview** menu entry is the same as the **Service Mesh** → **Overview** menu entry within the OpenShift console.

The **Traffic Graph** menu entry shows the same graph as the **Service Mesh** → **Traffic Graph** menu entry within the OpenShift console.

The **Workloads** menu entry information is the same as what you can see within the OpenShift console under the **Workloads** → **Deployments**, and other entries in the **Workloads** menu.

4.4. Click Applications

The only menu entry that you have in the external Kiali console, but not in the OpenShift console, is the **Applications** entry. This view shows the list of member elements, such as gateways and virtual services, classified by the `app=application_name` label in the workloads and services.

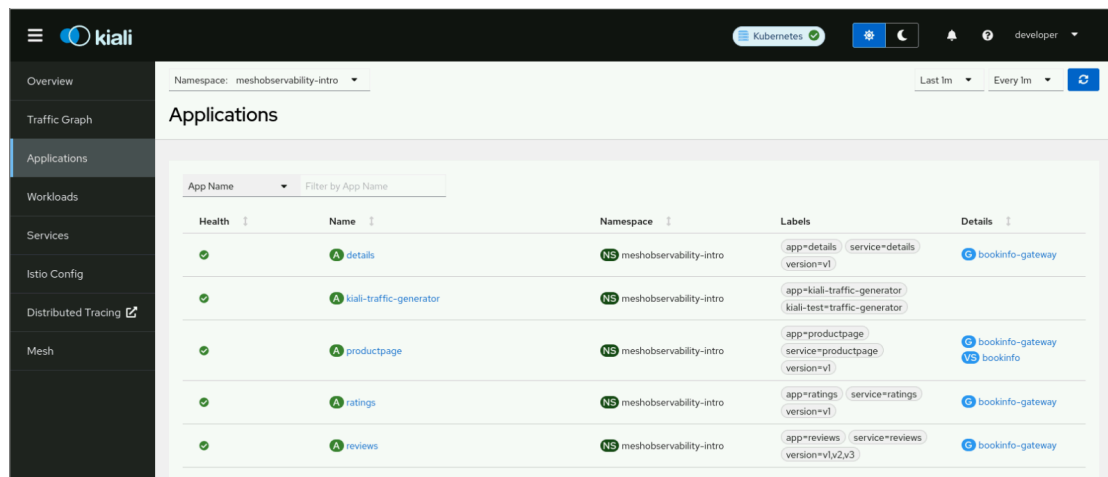


Figure 1.23: Kiali's applications menu

Finish

On the workstation machine, use the `lab` command to complete this exercise. This step is important to ensure that resources from previous exercises do not impact upcoming exercises.

```
[student@workstation ~]$ lab finish meshobservability-intro
```

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